

JOHN MERWIN

Generative AI Engineer

merwinj@hotmail.com | linkedin.com/in/john-merwin-ai | Washington, DC area | US Citizen

PROFESSIONAL SUMMARY

Generative AI Engineer with a long, continuous tenure at a multilateral development institution, spanning enterprise software engineering, business systems administration, and AI/ML delivery. Currently designing and deploying LLM-powered applications — including retrieval-augmented generation (RAG), multi-agent workflows, and semantic search — built on a foundation of over a decade of production software engineering across finance, nonprofit, and public-sector environments. Hands-on experience fine-tuning open-source LLMs (LoRA/PEFT) on multi-GPU infrastructure, alongside practical delivery experience across OpenAI, Anthropic Claude, and Google Gemini ecosystems.

CORE SKILLS

Generative AI & LLMs: RAG pipelines • Multi-agent systems (LangGraph, CrewAI) • Prompt engineering • Function calling & structured outputs • LangChain • LangSmith

Model Fine-Tuning: LoRA • PEFT • Hugging Face Transformers • Distributed training (PyTorch DDP) • BF16/FP16 precision optimization

LLM Platforms: OpenAI GPT • Anthropic Claude • Google Gemini • DeepSeek

Engineering & Data: Python • SQL • REST APIs • Vector databases (Pinecone, pgVector, ChromaDB) • Embeddings & semantic search • Pydantic

Machine Learning: Scikit-learn • XGBoost • Random Forest • Regression & classification • Feature engineering • Pandas • NumPy

Deployment & Cloud: Docker • AWS (EC2, Elastic Beanstalk, S3, CloudWatch) • Streamlit • Render • CI/CD pipelines

Analytics: Tableau • Power BI

PROFESSIONAL EXPERIENCE

Inter-American Development Bank — Washington, DC

Multilateral development bank, \$200B portfolio, 26 Latin American and Caribbean member countries. Continuous tenure spanning four roles.

Generative AI Engineer / Data Scientist

Oct 2023 – Present

- Design and deploy production RAG systems enabling natural language search across enterprise knowledge bases
- Build AI assistants using OpenAI, Claude, and Gemini APIs to automate research and documentation workflows
- Design multi-agent systems using LangGraph and CrewAI for multi-step reasoning and task orchestration
- Implement semantic search using vector databases (pgVector, Pinecone, ChromaDB)

Data Visualization Developer (Tableau/Power BI)

2021 – 2023

- Migrated the department's existing dashboard suite (originally built in Highcharts) to Tableau, then later to Power BI, as the Bank's standard BI tooling evolved
- Trained business users and improved adoption across teams

Salesforce Administrator

2019 – 2021

- Supported Salesforce rollout across 26 countries
- Automated business processes improving efficiency and data quality
- Earned 3 Salesforce certifications (Administrator track)

Software Engineer

2012 – 2019

- Maintained and extended legacy Classic ASP systems still in active use today, implementing new features to reflect evolving business rules
- Enhanced a VB.NET desktop application, also still in active use today — added automated outgoing email workflows and a cost calculation matrix
- Developed operational and management reports using SQL Server Reporting Services (SSRS)

- Led a department-wide initiative to design and build unit-specific dashboards using Highcharts, Bootstrap, AngularJS, C#, and JavaScript

INDEPENDENT PROJECTS

Self-directed technical projects built outside of employer scope, demonstrating hands-on depth across the modern GenAI stack.

DeepSeek 7B LoRA Fine-Tuning

LLM fine-tuning on distributed GPU infrastructure

- Fine-tuned DeepSeek-7B-Chat on the Alpaca instruction dataset (~52K examples) using LoRA/PEFT
- Configured distributed training across dual A100 80GB GPUs using PyTorch DDP (torchrun), resolving device-mapping conflicts between Accelerate hooks and multi-process training
- Applied BF16 precision and TF32 optimization to maximize GPU utilization and VRAM efficiency
- Stack: PyTorch 2.6, Hugging Face Transformers, PEFT, LoRA, Thunder Compute

Linguify — AI Document Translation App

- Built a provider-agnostic .docx translation app supporting English, French, Portuguese, and Spanish using Claude and GPT APIs
- Reduced document translation turnaround from 3-4 days to under 2 minutes per document avoiding an estimated \$1,500-1,800 in freelance translation costs per 12-page document (~6,000 words at \$0.25-0.30/word)
- Engineered parallel chunk translation (ThreadPoolExecutor, 4 workers) with exponential-backoff retry logic, cutting per-document translation time from days to minutes in testing
- Deployed on AWS Elastic Beanstalk (EC2, S3, CloudWatch, IAM/MFA) and Render with production hardening

EssayLens — Multi-Agent Essay Evaluator

- Built a multi-agent system evaluating essays across multiple dimensions in parallel, replacing single-sentence feedback with structured, multi-dimensional scoring
- Implemented structured outputs (Pydantic) for reliable scoring and a multi-stage orchestration pipeline; deployed via Streamlit
- Reduced essay evaluation time from an estimated 5 minutes (manual grading) to ~25 seconds, while expanding feedback from a single summary sentence to structured multi-dimensional scoring

Additional projects: a resume-to-job-description matching pipeline (parsing, gap analysis, and recommendations), and a multi-agent RAG research assistant with web-fallback retrieval and citation-based responses — details available on request.

MACHINE LEARNING COURSEWORK

Master's program in Data Science — applied supervised learning to real-world datasets:

- Built classification models (Logistic Regression, Random Forest, XGBoost) on MNIST digit recognition and breast cancer diagnostic datasets
- Built regression models on the California Housing dataset, exploring feature relationships and prediction accuracy

EDUCATION

Master of Science, Data Science — Summa Cum Laude — Eastern University, St. Davids, PA (2023)

Bachelor of Science, Psychology — Summa Cum Laude — George Mason University, Fairfax, VA (2010)

ADDITIONAL INFORMATION

Fluent in English, Hindi, and Urdu; working knowledge of Spanish

Portfolio and live project demos available upon request

Open to remote roles and hybrid opportunities in the Washington, DC area

Selected enterprise AI applications available for live demonstration during the interview process